

Simulation Output Governance Standard - (SOGS)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Simulation Output Governance Standard (SOGS) defines the governance conditions under which outputs produced by Simulation Execution are formally identified, classified, bounded, interpreted, documented, and connected to the execution basis that produced them so that the distinction between what the simulation directly produced, what is subsequently derived or inferred from that output, what limitations constrain its interpretation, and what the output does not establish remains explicit throughout downstream use.

SOGS governs the transition from:

Simulation Execution

to:

Governed Simulation Output

without allowing computational results to acquire meaning, certainty, applicability, or evidential strength that the Simulation Execution itself did not establish.

Core Question

What exactly did the simulation produce, how must Simulation Outputs be bounded and interpreted, and what may legitimately be inferred from them?

Operational Role

SOGS serves as the ninth Parent Standard of SIMULOS® and establishes the first governance layer after Simulation Execution.

The preceding standard:

SES — Simulation Execution Standard

establishes how the simulation was instantiated, controlled, observed, repeated, and documented.

SOGS asks the next architectural question:

What exactly emerged from that execution, what kind of output is it, what does it directly represent, and where must interpretation stop?

The governed progression is:

Governed Simulation Execution Record



Raw Simulation Output



Output Identification



Output Classification



Output Context Binding



Inference Governance



Output Limitation



Output Documentation



Governed Simulation Output



Reality Correlation Governed Modules 1. SOIM — Simulation Output Identification Module

Governs the identification, differentiation, origin binding, execution linkage, version relationship, and traceability of individual and aggregated Simulation Outputs. 2. SOCM — Simulation Output Classification Module

Governs how Simulation Outputs are classified according to their form, origin, processing state, aggregation, uncertainty context, and intended interpretive role. 3. SIM — Simulation Inference Module

Governs what may legitimately be inferred from Simulation Outputs and preserves the distinction between:

Observed Simulation Result

and:

Interpretive Inference. 4. SOLM — Simulation Output Limitation Module

Governs material limitations, applicability boundaries, unresolved conditions, uncertainty constraints, exclusions, and interpretive restrictions attached to Simulation Outputs. 5. SODM — Simulation Output Documentation Module

Governs the structured documentation, lineage, contextual binding, inference record, limitation record, and downstream traceability of governed Simulation Outputs.

Governance Flow

Simulation Execution

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SOIM — Output Identification

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SOCM — Output Classification

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SIM — Simulation Inference

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SOLM — Output Limitation

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SODM — Output Documentation

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Governed Simulation Output

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Reality Correlation Governance Governed Output Object

SOGS introduces the governed concept:

Simulation Output Object

A Simulation Output Object is an identifiable output or governed output collection produced from one or more Simulation Executions and preserved with sufficient execution context, classification, inference boundaries, limitations, and documentation to support legitimate downstream examination.

A Simulation Output Object may include: numerical result, state trajectory, event sequence, simulated failure, behavioral trace, probability distribution, performance measure, optimization result, agent action, decision path, generated scenario outcome, Monte Carlo distribution, visualization, aggregated execution result, statistical summary, or another identifiable product of Simulation Execution.

Direct Output

A Direct Simulation Output is produced directly by the Simulation Execution.

Examples include:

collision occurred

trajectory reached state X

temperature exceeded threshold Y

agent selected action Z

18 of 250,000 executions entered recovery failure

These are execution-derived outputs.

Derived Output

A Derived Simulation Output results from processing one or more Direct Simulation Outputs.

Examples include: averages, distributions, percentiles, rankings, aggregate failure rates, confidence estimates, transformed metrics, comparative measures, and post-processed visualizations.

Derived Outputs should remain traceable to their contributing Simulation Outputs and executions where material. Inference

An Inference is a conclusion drawn from one or more Simulation Outputs.

For example:

Output: 18 of 250,000 simulated executions entered recovery failure.

Inference: Recovery failure appears concentrated under a particular combination of simulated conditions.

These are not the same object. Output ≠ Inference Principle

SIMULOS® establishes:

A Simulation Output is what the simulation produced. An inference is what an

actor concludes from that output.

The two must remain distinguishable.

Inference ≠ Reality Principle

SIMULOS® further establishes:

An inference supported inside a simulation does not by itself establish that the same relationship exists in Operational Reality.

That transition belongs to downstream governance.

Output Context Binding

A Simulation Output should remain sufficiently connected to the execution basis that produced it.

Where material, this includes:

Simulation Object

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Purpose & Scope

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Reality Representation

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Assumptions

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Model & Environment

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Inputs & Data

↓

Scenario

↓

Execution

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Output

Without this context, an output may remain computationally correct while becoming methodologically misleading.

No Context-Free Output Principle

Simulation Outputs must not acquire broader meaning merely because their originating simulation context has been removed.

Output Classification

Outputs may require classification according to characteristics such as:

Direct / Derived

Individual / Aggregated

Deterministic / Stochastic

Expected / Unexpected

Nominal / Edge-Case / Stress

Complete / Partial

Final / Intermediate

Observed / Calculated

Comparative / Standalone

Execution-Specific / Campaign-Level

Classification enables downstream users to understand what type of Simulation Output they are examining.

Aggregated Simulation Output

An aggregated output may combine many executions.

For example:

2,000,000 Monte Carlo runs → Failure Probability Estimate

SOGS requires the aggregate to remain sufficiently connected to: contributing execution population, relevant exclusions, failed runs, scenario population, repetition structure, and aggregation method.

No Aggregation Concealment Principle

Aggregation must not erase materially relevant execution behavior that changes interpretation of the aggregated result.

Output Population

A Simulation Output may represent:

one execution

one Repetition Set

one Scenario

multiple scenarios

one campaign

or:

multiple simulation populations.

The population basis should remain identifiable.

Output Exclusion

Where executions or results are excluded from an output population, material exclusions should remain visible.

This prevents: silent failed-run removal, cherry-picking, survivor-only reporting, post hoc filtering, and misleading aggregate construction.

Output Limitation

Every materially relevant Simulation Output may carry limitations arising from: assumptions, abstraction, fidelity, model boundaries, environment boundaries, input uncertainty, data representativeness, scenario coverage, execution conditions, repetition sufficiency, observation limitations, output processing, or unresolved simulation behavior.

SOGS does not require all limitations to invalidate the output.

It requires them to remain visible where they materially constrain interpretation.

Inherited Limitation Principle

A Simulation Output inherits materially relevant limitations from the simulation conditions upon which it depends.

Post-processing does not erase those limitations.

No Precision Inflation Principle

A highly precise numerical output does not establish equivalent certainty about reality.

For example:

Failure Probability = 0.003742%

may be computationally precise while remaining highly sensitive to: assumptions, scenario selection, model fidelity, data quality, and representation boundaries.

Computational Precision ≠ Epistemic Certainty

SIMULOS® establishes:

The numerical precision of a Simulation Output must not be confused with the strength of what that output legitimately establishes.

Unexpected Output

A Simulation Output may reveal behavior not anticipated by the Simulation Purpose or Scenario Design.

Unexpected output should not automatically be: deleted, classified as error, excluded, or normalized away.

It may reveal: hidden interaction, model instability, unmodeled dependency, scenario insufficiency, rare behavior, or a new simulation question.

Output-to-Scenario Feedback

A materially unexpected output may therefore create:

New Scenario Requirement

or:

Scenario Expansion Requirement

under SSS.

SIMULOS® is therefore capable of recursive simulation learning.

Output-to-Assumption Feedback

An output may expose sensitivity to an assumption.

This may create:

Assumption Reassessment

under SAS.

Output-to-Model Feedback

Unexpected or unstable outputs may create a need to reassess:

Simulation Model & Environment

under SMES.

Output-to-Input Feedback

Output behavior may expose input sensitivity, missing data regions, or representativeness problems requiring reassessment under SIDS.

Output-to-Execution Feedback

An output anomaly may originate from: execution deviation, hidden state, incomplete run, control intervention, or repetition drift.

SOGS therefore preserves backward traceability into SES.

No Automatic Failure Attribution

An unexpected Simulation Output does not establish that the simulated system failed, the simulation failed, or Operational Reality would fail.

Those remain separate questions requiring appropriate governance.

Inference Boundary

An inference should remain bounded by: Simulation Purpose, Simulation Scope, represented reality, assumptions, model and environment, inputs, scenario population, execution population, output population, uncertainty, and known limitations.

Inference Escalation

SOGS distinguishes between increasingly strong claims.

For example:

Level 1 — Execution Observation

“Failure occurred in Run 184.”

Level 2 — Repetition Observation

“Failure occurred in 18 of 250,000 repetitions.”

Level 3 — Simulation Pattern Inference

“Failure is associated with simulated low-power recovery conditions.”

Level 4 — Reality-Oriented Claim

“The real spacecraft will fail under low-power recovery conditions.”

The first three may potentially be governed within simulation output analysis.

The fourth crosses toward:

Reality Correlation

and cannot be established by SOGS alone.

No Reality Leap Principle

SIMULOS® establishes:

A Simulation Output must not move directly from simulated behavior to a claim about Operational Reality without passing through the appropriate reality-correlation and validation governance.

Output Documentation

A governed Simulation Output should preserve, where material: Output ID, Output Type, originating Execution ID, Repetition Set, Scenario ID, model version, environment version, Input Set, relevant parameters, output population, aggregation method, processing history, inference status, limitations, uncertainty context, exclusions, and downstream use.

Output Lineage

Output lineage may follow:

Execution

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Direct Output

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Derived Output

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Aggregated Output

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Inference

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Downstream Assessment

This allows later reconstruction of how a conclusion emerged from Simulation Execution.

No Output Laundering Principle

SIMULOS® establishes:

A Derived Output or inference must not acquire stronger methodological status merely because its relationship to the original Simulation Execution has become less visible.

Output Transformation

Outputs may undergo: filtering, normalization, aggregation, smoothing, dimensional reduction, statistical transformation, visualization, AI summarization, or other processing.

Material transformations should remain traceable where they affect interpretation.

AI Interpretation of Simulation Outputs

AI systems may analyze enormous Simulation Output populations.

They may: identify patterns, detect anomalies, cluster failures, generate summaries, propose causal hypotheses, or identify new scenarios.

SOGS distinguishes:

Simulation Output

from:

AI-Generated Interpretation of Simulation Output.

AI-Generated Inference Principle

An AI-generated interpretation does not become a Simulation Output merely because it was generated automatically from Simulation Outputs.

Its interpretive status should remain visible.

Output Conflict

Different executions or Simulation Output populations may support conflicting interpretations.

SOGS preserves conflict rather than forcing artificial agreement.

Possible status:

Output Consistent

Output Conditionally Consistent

Output Mixed

Output Conflicted

Output Relationship Undetermined Output Uncertainty

Output uncertainty may arise from multiple sources.

SOGS may preserve:

Input Uncertainty

Parameter Uncertainty

Stochastic Uncertainty

Model Uncertainty

Scenario Uncertainty

Execution Uncertainty

Observation Uncertainty

and:

Inference Uncertainty.

These should not automatically be collapsed into one generic uncertainty number.

Output Sufficiency

A Simulation Output may be technically valid as an output while still insufficient for a particular downstream inference.

Therefore:

Output Existence $\neq$ Inference Sufficiency.

Simulation Campaign Bottleneck

Modern simulation systems can produce more outputs than organizations can meaningfully interpret.

The governance bottleneck may therefore shift from:

Can we generate simulation results?

to:

Can we determine what those results actually mean, what they do not mean, and which decisions they may legitimately support?

SOGS governs this transition.

Space & Mission Simulation Application

A space organization may run millions of mission simulations producing: trajectory distributions, navigation errors, failure modes, communication losses, recovery outcomes, fuel margins, and mission-success estimates.

The critical governance question is not merely:

What percentage succeeded?

It is also:

Which execution population produced that percentage?

Which scenarios were included?

Which failed executions were excluded?

Which assumptions materially shaped the result?

What does the output directly establish?

Which conclusions are interpretations?

Which conclusions require correlation with Operational Reality?

SOGS provides the structure required to answer these questions.

Autonomous Systems Application

An autonomous robot may complete:

99.97% of simulated tasks successfully.

SOGS prevents this result from automatically becoming:

The robot is 99.97% safe in reality.

The simulation output may legitimately establish a performance property within the governed simulation conditions.

Moving from that result toward real-world safety requires downstream governance.

Relationship to SES

SES governs:

how Simulation Execution occurs.

SOGS governs:

what emerges from that execution and how that output may be interpreted.

The formal handoff is:

Governed Simulation Execution Record



Simulation Output Object Relationship to RRS

RRS governs how reality is represented inside the simulation.

SOGS preserves relevant representation limitations when interpreting outputs.

Relationship to SAS

SAS governs Simulation Assumptions.

SOGS ensures materially relevant assumption limitations propagate into output interpretation.

Relationship to SSS

SSS governs Scenario Design and Coverage.

SOGS ensures outputs are not interpreted beyond the scenario population that produced them.

Relationship to ORA™

ORA™ governs actual Operational Reality.

SOGS does not determine whether a Simulation Output corresponds to reality.

That question belongs to the downstream reality-correlation governance space.

Relationship to VALIDOS®

VALIDOS® determines whether a Simulation Output, inference, correlation claim, simulation methodology, or other Validation Object can legitimately support a formal validation claim.

SOGS governs the Simulation Output before that validation determination.

Relationship to INTEGROS®

SOGS deliberately does not govern general Output Integrity.

General integrity remains under INTEGROS®.

SOGS governs:

identification, classification, inference, limitation, and documentation of Simulation Outputs.

Relationship to ADIT®

ADIT® may audit Simulation Output records, inference pathways, exclusions, and traceability.

SOGS governs the output space itself.

Governance Boundary

SOGS begins when one or more identifiable outputs emerge from a governed Simulation Execution.

SOGS ends when those outputs have been sufficiently: identified, execution-bound, classified, population-bounded, inference-bounded, limitation-bound, transformation-traced, and documented

to form a Governed Simulation Output suitable for downstream Reality Correlation.

SOGS does not determine: whether the simulation accurately represents Operational Reality, whether a Simulation Output is externally valid, whether the simulated behavior will occur in reality, whether a system is operationally safe, whether a Simulation Output constitutes sufficient Validation Evidence, or whether operational reliance is justified.

Minimum Implementation Framework 1.

Identify the Simulation Output

Assign a stable Output ID and establish what exactly was produced. 2. Bind the Output to Execution

Connect the output to its originating Simulation Execution, Repetition Set, or Campaign. 3. Classify the Output

Determine whether the output is: direct, derived, aggregated, deterministic, stochastic, intermediate, final, or another governed class.

4. Establish the Output Population

Identify which executions and scenarios contribute to the output. 5. Preserve Material Exclusions

Document excluded executions, failed runs, filtered results, and other material population changes. 6. Trace Material Transformations

Preserve transformations affecting output interpretation. 7. Separate Output From Inference

Distinguish what the simulation produced from conclusions subsequently drawn from it. 8. Bound the Inference

Ensure inference does not exceed the simulation conditions supporting it. 9. Propagate Material Limitations

Carry forward relevant limitations from upstream simulation governance. 10. Assess Output Uncertainty Context

Identify material uncertainty affecting interpretation. 11. Establish Output Documentation

Preserve the context, lineage, inference, limitations, and downstream relationships. 12. Establish the Governed Simulation Output

Create the formal output object suitable for Reality Correlation and downstream validation.

Governance Outputs

SOGS may produce: Simulation Output ID, Simulation Output Object, Direct Simulation Output Record, Derived Simulation Output Record, Aggregated Simulation Output Record, Output Classification, Output Population Record, Output Exclusion Record, Output Transformation Record, Output-to-Execution Traceability Record, Output Lineage Record, Simulation Inference Record, Inference Classification, Inference Boundary, Inference Limitation Record, Output Limitation Record, Inherited Limitation Record, Output Uncertainty Record, Output Conflict Record, Output Sufficiency Assessment, AI-Generated Interpretation Record, Output Documentation Record, Reality-Correlation Handoff Record, and Governed Simulation Output.

Architectural Position

Simulation Output Governance Standard (SOGS) is the ninth Parent Standard of SIMULOS®.

Its position is:

8. Simulation Execution Standard (SES)

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9. Simulation Output Governance Standard (SOGS)

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10. Reality Correlation Governance

SOGS therefore forms the critical boundary between:

what happened inside the simulation

and:

what that simulated result may legitimately tell us about something beyond the simulation.

Foundational Principle

A Simulation Output becomes governable only when what the simulation directly produced remains distinguishable from how that output was processed, interpreted, limited, aggregated, and ultimately used to support claims beyond the execution that produced it.

Canonical Closing Statement

Simulation Output Governance Standard establishes the governed transition from Simulation Execution to interpretable Simulation Output by defining how outputs are identified, classified, execution-bound, population-bounded, transformed, inferred from, limitation-bound, uncertainty-contextualized, and documented. Through explicit separation of direct output, derived output, aggregation, and inference, SOGS prevents computational precision from becoming false certainty, simulation behavior from becoming an unsupported claim about Operational Reality, and large-scale simulation results from losing the execution context and limitations necessary for legitimate interpretation. SOGS therefore creates the governed Simulation Output Object that can move forward into Reality Correlation and downstream validation without exceeding what the simulation itself actually established.

Module Architecture

- [Simulation Output Identification Module \(SOIM\)](#)
 - [Simulation Execution Control Module \(SECM\)](#)
 - [Simulation Observation Module \(SOM\)](#)
 - [Simulation Output Limitation Module \(SOLM\)](#)
 - [Simulation Output Documentation Module \(SODM\)](#)
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Related Documents

- [About Simulation Output Governance Standard \(SOGS\)](#)
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